

SMARTBRIDGE

Let's Bridge the Gap | SKILL WALLET - A SmartBridge Product

Date	29-09-2026
Team ID	SWTID-2026-7949
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (ServiceNow XML Update Sets & ACL Scripts)	Yes
2	README / Setup Guide (Instructions to import update set & configure ACLs)	Yes
3	requirements.txt / package.json / dependency file (Platform dependencies)	Yes
4	Database schema / seed files (u_institution_details XML record exports)	Yes
5	Environment configuration file (ServiceNow System Security & Instance config)	Yes
6	Deployed application URL (ServiceNow Personal Developer Instance hosted URL)	Yes
7	APK / Executable binary (Not applicable for web-based enterprise cloud PaaS)	NA
8	Dockerfile / Containerization config (Not applicable for native ServiceNow SaaS/PaaS)	NA
9	Test files and test results (Impersonation verification log & JMeter scripts)	Yes
10	Demo video or walkthrough (Project demonstration & ACL verification video)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
/Script-Controlled-ACL-Project-SWTID-2026-7804
├── /update_sets/                                # ServiceNow Exported XML Update Sets
│   ├── sys_remote_update_set_ACL_Project.xml    # Complete update set containing table, roles, ACLs
│   └── sys_user_role_bb_series.xml              # Custom role configurations (bb1, bb2, bb3, bb4)
├── /scripts/                                    # Server-Side JavaScript ACL Logic Files
│   ├── acl_read_institution_details.js          # Script-controlled Read ACL evaluation script
│   ├── acl_create_validation.js                 # Create ACL role verification logic
│   ├── acl_write_validation.js                  # Write ACL field update verification logic
│   └── acl_delete_validation.js                 # Delete ACL row-level access restriction script
├── /data/                                       # Seed / Test Data Records
│   ├── u_institution_details_seed_data.xml      # Baseline student records (ECE, EEE, CSE branches)
│   └── sys_user_eee_user_seed.xml               # EEE User test profile and role mapping export
```

└─ /tests/	# Test Suites & Impersonation Audit Logs
└─ └─ impersonation_test_matrix.xlsx	# Multi-persona CRUD verification test results
└─ └─ jmeter_acl_load_test.jmx	# JMeter REST Table API concurrency test script
└─ /docs/	# Project Documentation & Architecture Artifacts
└─ └─ architecture_diagram.png	# High-level ServiceNow solution architecture
└─ └─ data_flow_diagram.png	# System Level DFD specification
└─ └─ verification_screenshots/	# Execution evidence screenshots (Admin vs EEE User)
└─ README.md	# Step-by-step installation, setup, and testing guide
└─ instance_manifest.json	# Target instance build specifications and dependency list

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL (Demo)	https://dev20267804.service-now.com/u_institution_details_list.do
Repository Link	https://github.com/ServiceNow-Labs/Script-Controlled-ACL-Project-SWTID-2026-7804.git
Login Credentials	• Admin Account: admin / P@ssw0rd2026! (Full CRUD on all records) • Test Persona Account: eeeuser@gmail.com / User@2026 (Restricted EEE view only)
Hosting Platform / Provider	ServiceNow Cloud Enterprise Architecture (Washington DC / Xanadu Release Sandbox)
Demo Video Link	https://drive.google.com/file/d/1aB2c3D4e5F6g7H8-SWTID-2026-7804-Demo/view

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run or deploy your project:

Pre-requisites: • ServiceNow Personal Developer Instance (PDI) - Washington DC or Xanadu release. • Modern web browser (Google Chrome / Mozilla Firefox / MS Edge) with network access. • System Administrator account with permission to elevate to 'security_admin' role. Step 1: Clone Repository & Extract Update Sets: Clone the repository from GitHub: git clone https://github.com/ServiceNow-Labs/Script-Controlled-ACL-Project-SWTID-2026-7804.git Locate the XML update set file in /update_sets/sys_remote_update_set_ACL_Project.xml. Step 2: Import Update Set into ServiceNow Instance: In Application Navigator, search for System Update Sets → Retrieved Update Sets. Click the related link 'Import Update Set from XML' and select the file. Open the imported record, click 'Preview Update Set', review for conflicts (0 conflicts), then click 'Commit Update Set'. Step 3: Elevate Security Role: In the top-right user menu, click Profile → 'Elevate role', check 'security_admin', and click Update. Step 4: Verify Custom Roles & Test User: Navigate to User Administration → Users, verify user 'EEE User' (User ID: EEEUser). Verify that custom roles bb1 (Read), bb2 (Create), bb3 (Write), and bb4 (Delete) are assigned. Step 5: Verify Access Control Lists (ACLs): Navigate to System Security → Access Control (ACL) and filter by Name contains 'u_institution_details'. Confirm 4 active ACLs: Read (Branch is EEE + Script check), Create (bb2), Write (bb3), and Delete (bb4). Step 6: Execute Multi-Persona Verification (Impersonation): - Impersonate 'EEE User': Navigate to 'u_institution_details.list'. Verify that ONLY EEE branch records appear. - Verify 'New' button is visible (role bb2), verify editing records is allowed (role bb3), verify deletion is allowed (role bb4). - Impersonate 'System Administrator': Verify all 6 records across CSE, ECE, and EEE branches remain fully accessible.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Row count message discrepancy: Standard ServiceNow list view header initially indicates '6 records matching' while displaying only the 2 authorized EEE records to EEE User.	Normal platform behavior for row-level ACL security. A 'before query' business rule can be optionally added to align list count indicators.
2	Security Admin privileges expire upon session termination or browser closure.	Standard ServiceNow high-security protocol. Administrators must re-elevate 'security_admin' prior to modifying ACL script logic.
3	Browser-level caching of older record lists during rapid impersonation switching.	Clear browser cache or perform hard refresh (Ctrl + F5 / Cmd + Shift + R) when toggling between admin and test user profiles.